

Microplastic Degradation Research Project

Phase 4 and 5 — Codon Optimization and Cloning Construct Assembly

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Phase 4 converts the engineered protein sequence into *E. coli*-optimized DNA using the Sharp and Li (1987) Codon Adaptation Index. Phase 5 assembles the full NdeI/6xHis/XhoI cloning constructs ready for gene synthesis.

Phase 4 — Codon Optimization Results

Property	Engineered Variant	Wild-type Control
Protein length (aa)	290	290
DNA length (bp)	870	870
GC content (%)	56.67	57.47
CAI	1.000	1.000
Back-translation verified	OK	OK

CAI of 1.000 is the theoretical maximum using Sharp and Li (1987) *E. coli* K-12 optimal codons. GC content difference (0.80%) directly reflects the five Arg to Glu mutations — each swaps CGT (2 GC) for GAA (1 GC).

Phase 5 — Cloning Construct Results

Check	Engineered	Wild-type
Insert length (bp)	903	903
GC content (%)	56.59	57.36
N-terminal sequence	MHHHHHHMNF...	MHHHHHHMNF...
His-tag confirmed	YES	YES
Full protein length (aa)	297	297
Internal NdeI site	None	None
Internal XhoI site	None	None
Back-translation verified	OK	OK
Ready for gene synthesis	YES	YES

903 bp = 6 (NdeI) + 18 (His6 tag) + 870 (CDS) + 3 (stop) + 6 (XhoI). Both constructs verified clean — ready to submit to Twist Bioscience, IDT, or Genscript. Expression host: *E. coli* SHuffle T7.

Project Status

Phase	Description	Status
Phase 1	Pepsin cleavage site mapping	COMPLETE
Phase 2	Structural SASA analysis	COMPLETE
Phase 3	8-mutation variant design	COMPLETE
Phase 4	Codon optimization CAI=1.000	COMPLETE
Phase 5	903 bp cloning constructs assembled	COMPLETE
Phase 6	Molecular dynamics simulations	PENDING